

# FINAL EXAM REVIEW

10-701: INTRODUCTION TO MACHINE LEARNING

11/26/2024

## 1 Neural Networks & Backpropagation

1. Your friend wants to build a fully-connected one-hidden-layer Neural Network. Her inputs have 9 features. She wants 7 hidden neurons in the first layer and 2 neurons in the output layer generated using a softmax function. Each layer is a linear layer fully-connected to the previous one and includes a bias term. Each scalar counts as one parameter. Both hidden layers have RELU activation function.

- (a) What is the number of parameters she will need to create this neural network? Each scalar counts as one parameter.

- (b) Your friend shares with you that the output she is trying to predict is binary. Using this information, describe an architecture that can learn the same decision boundary while reducing the number of parameters.

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2. Your friend trains a Neural Network on some training dataset and achieves an almost perfect training accuracy. However they perform very poorly when it comes to test time. In the boxes below, write either **{Increase, Decrease or N/A}** to indicate what they should do with their model to reduce the amount of overfitting (N/A here means changing the specified quantity won't affect overfitting). *You should assume the training and test data are representative of the true underlying distribution of the data.*

the number of training data

the number of test data

the regularization constant

the number of nodes in the output layer

3. In backpropagation, why is it necessary that we loop through the layers in reverse-order, moving from the output layer backwards towards the input layer?

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4. When training a neural network, which of the following techniques is/are *primarily* used to address the nonconvexity of the loss function?

- ☐ Weight decay
- ☐ Random restarts
- ☐ Early termination
- ☐ None of the above

5. **Short Answer:** Calculate the gradient of a sigmoid function, which is defined as  $\sigma(x) = \frac{1}{1+e^{-x}}$ . Show your work.

6. **Short Answer:** Based on the previous answer, how can a sigmoid contribute to the vanishing gradient problem?

## 2 Deep Learning, CNNs & RNNs

1. What is the *primary* purpose of a convolutional layer in a CNN?

- ☐ To reduce the dimensions of the input image.
- ☐ To detect features such as edges and textures in the input image.
- ☐ To classify the input image into various categories.
- ☐ To flatten the input image across channels.

2. Suppose you are building a CNN that takes input images with 3 channels, each of size  $20 \times 20$  pixels. Your first convolutional layer takes the 3 input channels and produces 5 output channels; it uses  $4 \times 4$  filters. It also uses a padding of 2 (along all sides of the image) and a stride of 2 (in both dimensions).

(a) How many parameters are there in this convolutional layer, including the bias terms?

(b) What is the dimensionality of the output of this convolutional layer? Include the channel dimension in your answer and make sure you clearly indicate which dimension is the channel dimension.

3. Which of the following statements about RNNs is/are correct?

- ☐ Training RNNs is difficult because of vanishing and/or exploding gradients.
- ☐ Gradient clipping is an effective technique to address the vanishing gradient problem.
- ☐ RNNs differ from feed-forward neural networks in that they have an additional weight matrix connecting hidden layers across time-steps.
- ☐ RNNs can process sequences of arbitrary length, while feed-forward neural networks can not.
- ☐ None of the above.

4. Briefly describe the bidirectional RNN architecture and describe why it is *not* appropriate for language modelling.

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5. **Multiple Choice:** What is the time complexity of softmax self-attention as a function of the input length  $n$ ?

- ☐  $O(\log n)$
- ☐  $O(n)$
- ☐  $O(n \log n)$
- ☐  $O(n^2)$
- ☐  $O(n^3)$

6. **Short Answer:** Justify your choice from above.

### 3 Attention & Transformers

1. For a fixed input-size and embedding dimension, which of the following statements is *not* true about multi-head attention relative to single-head attention?

- ☐ Multi-head attention is more suitable for parallel computation than single-head attention.
- ☐ Multi-head attention layers have more total parameters in their query, key and value matrices than single-head attention layers.
- ☐ Multi-head attention is compatible with scaled dot-product attention while single-head attention is not.
- ☐ Multi-head attention are able to capture more diverse relationships between tokens than single-head attention.
- ☐ None of the above

2. Suppose you have a transformer model that employs multi-head attention. The inputs to your model are sequences of  $T$  tokens and each token is represented by a  $d_M$ -dimensional embedding. Your model has  $H$  heads and for each head, the dimensionality of the key and query vectors is  $d_K$  and the dimensionality of the output vectors is  $d_V$ .

- (a) What are the dimensions of the key matrix *for one of the attention heads* i.e.,  $K^h$  where  $K^h = XW^h$ ?

- (b) What is the dimensionality of the multi-headed attention output *before any sort of concatenation*?

3. Briefly explain why positional encodings are used in transformer models.

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## 4 Clustering

1. **Select all that apply:** Which of the following are benefits of hierarchical clustering?
  - ☐ Does not require specifying the number of clusters in advance
  - ☐ Captures nested or hierarchical relationships within the data
  - ☐ Interpretable results for how clusters are related and formed
  - ☐ None of the above
2. **select One:** We are given a dataset of  $n$  samples,  $x_1, \dots, x_n$  where  $x \in \mathbb{R}^2$  and asked to run k-means. If we choose the value for  $k$  to optimize the objective function how many clusters will be used (i.e. what value of  $k$  will we choose)?
  - ☐ 1
  - ☐ 2
  - ☐  $n$
  - ☐  $n^2$
3. **select One:** Why do we use Lloyd's algorithm to compute the K-means solution instead of the brute force solution?
  - ☐ Lloyd's algorithm is guaranteed to find a solution.
  - ☐ Lloyd's algorithm is guaranteed to converge in  $O(n)$  steps.
  - ☐ The brute force solution is NP hard.
  - ☐ The brute force solution is a polynomial time algorithm
4. We are given data drawn from two independent multivariate Gaussian distributions. The data are shown in Figure 1 with the initial cluster centers as red crosses.

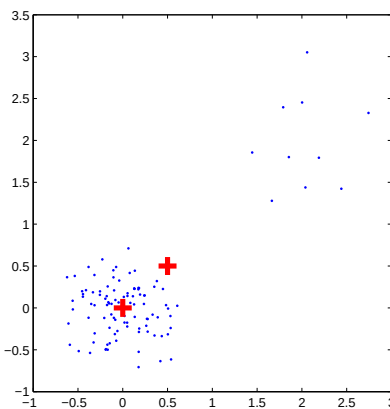
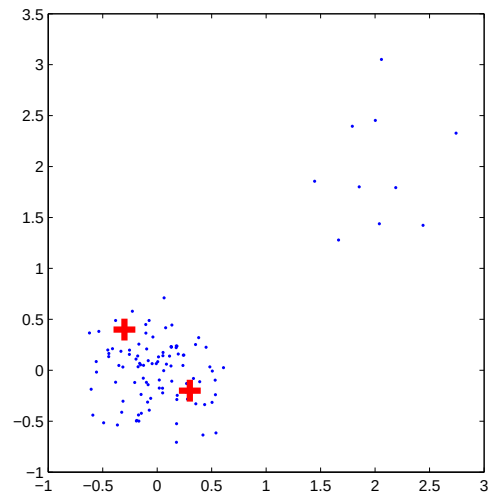
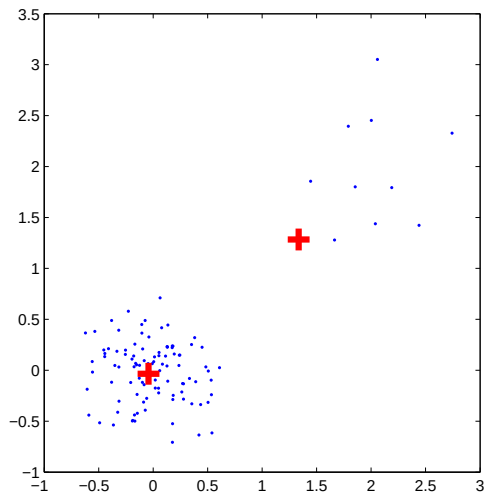
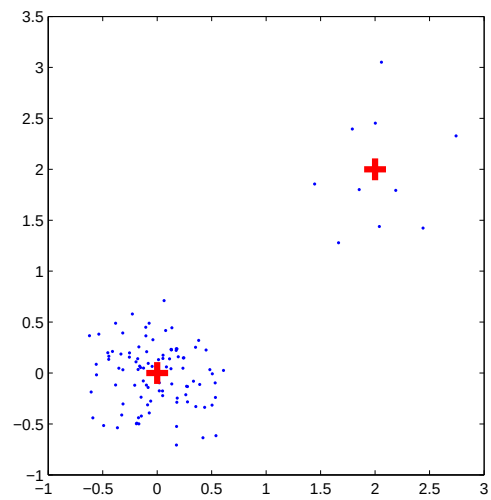
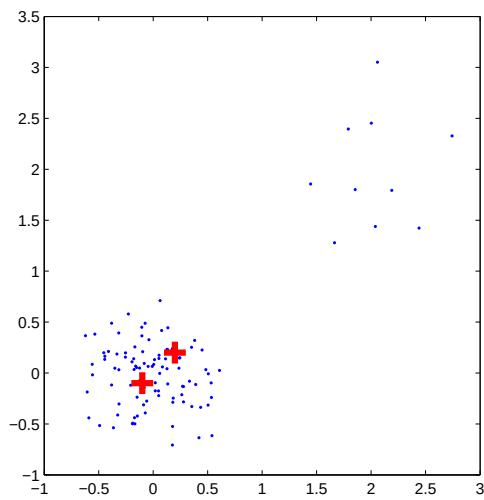


Figure 1: Initial data and cluster centers

- (a) Circle the image which depicts the cluster center positions after 1 iteration of Lloyd's algorithm.



5. The TAs for 10-701 pulled an all nighter and designed a brand new initialization algorithm to initialize the cluster centers for Lloyd's method. Below is the algorithm which they called "K-means %%".

- Randomly choose the first cluster center,  $c_1$
- For  $j = 2 : k$  iterate
  - Choose  $c_j$  to be one of the data points according to the following weighted probability:

$$P(c_j = x) \propto \min_{l < m} \frac{1}{\|x - c_l\|}$$

Assume that  $x \in \mathbb{R}$ . Answer the following questions about "K-means %%":

- (a) Assume we have three data points  $(x = 3), (x = 1), (x = -2)$  and we set  $k = 2$ . If we choose the first cluster center to be  $c_1 = 2$ , what is the probability the we choose  $c_2$  to be the following values:

$$P(c_2 = -2) =$$

$$P(c_2 = -1) =$$

$$P(c_2 = 1) =$$

$$P(c_2 = 3) =$$

6. Is "K-means %%" an effective initialization algorithm?

- ☐ Yes  
☐ No

7. Explain your answer to part (b).

8. **Select all that apply:** Which of the following statements is true about clustering using Lloyd's algorithm?

- ☐ Lloyd's algorithm always converges to a global optimum.  
☐ The choice of initialization greatly affects the performance of the algorithm.  
☐ With random initialization, increasing the selected number of clusters  $k$  makes it more likely that the initialization is better.  
☐ Lloyd's method is proved to converge only if initialized with k-means++.



9. **Select all that apply:** Say you want to use k-means clustering to partition  $N$  data points, where  $1 < k < N$ . Which of the following is a sensible way to pick the value of  $k$ ?
- ☐ Choose  $k$  such that the  $k$ -means cost function is minimized on held out data.
  - ☐ Choose  $k$  such that  $k, k + 1, \dots, N$  have similar values of the cost function but that  $1, \dots, k - 1$  have much higher costs, where all costs are evaluated from training data.
  - ☐ Set  $k = \frac{n}{2}$
  - ☐ Set  $k = \log(\frac{n}{2})$

10. Consider a data set of English words, in which all words have the same length. The distance measure between pairs of data points (words) is defined as the number of substitutions in letters required to transform one string to another. This is a modified version of the edit distance<sup>1</sup> between the two strings. For example, the distance between **baba** and **bebe** is 2, since we have to substitute two letters (**a** and **a** with **e** and **e**) in the first word to transform it into the second.

Consider the following words:

{litter, lister, kitten, mitten, bitter, booted}

These words have the following modified edit distances in the distance matrix:

|        | litter | lister | kitten | mitten | bitter | booted |
|--------|--------|--------|--------|--------|--------|--------|
| litter | 0      | 1      | 2      | 2      | 1      | 4      |
| lister | 1      | 0      | 3      | 3      | 2      | 4      |
| kitten | 2      | 3      | 0      | 1      | 2      | 4      |
| mitten | 2      | 3      | 1      | 0      | 2      | 4      |
| bitter | 1      | 2      | 2      | 2      | 0      | 3      |
| booted | 4      | 4      | 4      | 4      | 3      | 0      |

We now perform bottom-up hierarchical clustering using single or complete linkage, and at an intermediate stage, we have the following three clusters:

**C1** {litter, lister}

**C2** {kitten, mitten}

**C3** {bitter, booted}

- (a) [6 pts] **Select One:** Which cluster will be merged with **C1** in the next iteration? **Circle the correct option, and then justify your answer.**

- ☐ **C2** with either single or complete linkage
- ☐ **C3** with either single or complete linkage
- ☐ **C2** with single-linkage or **C3** with complete-linkage
- ☐ **C3** with single-linkage or **C2** with complete-linkage

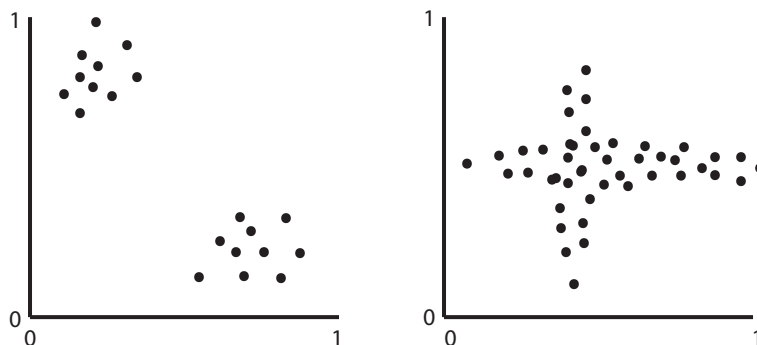
**Justification:**

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<sup>1</sup>While edit distance typically includes additions, deletions and substitutions as the possible edit operations, in this problem we count only substitutions.

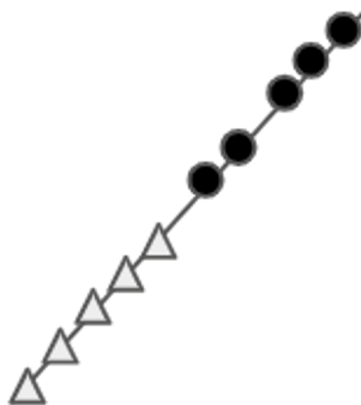
## 5 Dimensionality Reduction

1. **Drawing:** Consider the following two datasets. On the figures below, draw arrows from the mean of the data to denote the direction and relative magnitudes of the principal components.



2. Suppose the points in a dataset lie in the 2-D Euclidean space, and the set of labels is  $\mathcal{Y} = \{1, -1\}$ . The dataset contains a finite number of points drawn according to some distribution  $\mathcal{D}$ . For each of the following distributions  $D_i$ , determine if computing the principle components is likely to be helpful in classification. If your answer is yes, explain how this is useful.

If your answer is no, give a short reason why. In either case your answer should be 1-2 lines.



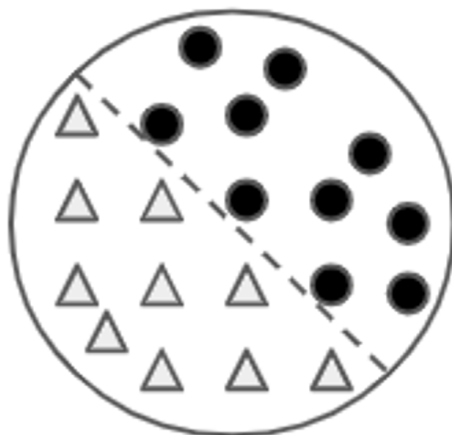
i.  $\mathcal{D}_1$

- (a) **Short Answer:**  $\mathcal{D}_1$ : Points  $(x_1, x_2)$  are drawn uniformly from the line segment joining points  $(0, 0)$  and  $(2, 2)$ . The label is 1 if  $x_2 > 1$ ,  $-1$  otherwise.



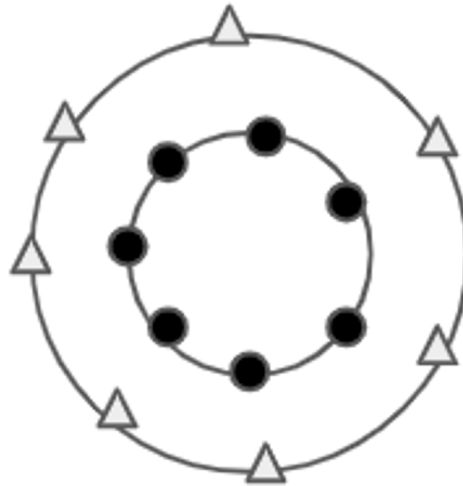
ii.  $\mathcal{D}_2$

- (b) **Short Answer:**  $\mathcal{D}_2$ : Points  $(x_1, x_2)$  are drawn uniformly from the line segments  $L_1$  and  $L_2$ ,  $L_1$  joins points  $(0, 1)$  and  $(1, 0)$  and  $L_2$  joins points  $(0.1, 1.1)$  and  $(1.1, 0.1)$ . The label is 1 if  $x_1 + x_2 = 1$ ,  $-1$  otherwise.



iii.  $\mathcal{D}_3$

- (c) **Short Answer:**  $\mathcal{D}_3$ : Points  $(x_1, x_2)$  are drawn uniformly inside the circle  $x_1 + x_2 \leq 2$ . The label is 1 if  $x_1 + x_2 \geq 0$ ,  $-1$  otherwise.



iv.  $\mathcal{D}_4$

- (d) **Short Answer:**  $\mathcal{D}_4$ : Points  $(x_1, x_2)$  are drawn uniformly from concentric circles  $C_1$  and  $C_2$  centers at  $(0, 0)$ ,  $C_1$  has radius 1,  $C_2$  has radius 2. The label is 1 if  $x_1^2 + x_2^2 = 1$ ,  $-1$  otherwise.

3. Answer each of the following questions with T or F. No justification required.

- (a) T or F The goal of PCA is to interpret the underlying structure of the data in terms of the principal components that are best at predicting the output variable.
- (b) T or F The output of PCA is a new representation of the data that is always of lower dimensionality than the original feature representation.
- (c) T or F The output of PCA is ordered according to the amount of variance each new feature contains.
- (d) T or F Subsequent principal components are orthogonal to each other.
- (e) T or F The output of PCA is at least as good as the original feature representation when the task we are trying to solve is unsupervised.

## 6 Reinforcement Learning

1. **Short answer:** Your friend, Ronald Weasley, is using reinforcement learning to learn an optimal policy on the following environment:

- There are two states ‘**In Play**’, ‘**Game Over**’. The agent always starts in the state ‘**In Play**’, and the state ‘**Game Over**’ is terminal, so if the agent gets there, the trajectory terminates.
- In the state ‘**In Play**’ there are two actions which the agent can take:
  - ‘**Score Goal**’: this action returns the agent to the ‘**In Play**’ state and gives the agent +10 reward.
  - ‘**Capture Golden Snitch**’: this action sends the agent to the ‘**Game Over**’ state and gives the agent +150 reward.
- The discount factor is  $\gamma = 1$ .

Ronald is excited about his model, however after looking over it, your other friend, Hermione Granger, says that reinforcement learning won’t work as Ronald intends.

- (a) **Short Answer:** What is the optimal policy  $\pi$ ? Explain why this is the case and provide  $V_\pi(\text{‘In Play’})$ .

- (b) **Math Answer:** Provide an alternative value of  $R(s = \text{‘In Play’}, a = \text{‘Score Goal’})$  that will lead to a different optimal policy  $\pi$ . Assume all other aspect of the environment remain as described in the problem statement.

- (c) **Math Answer:** Provide an alternative value of  $\gamma$  that will lead to a different optimal policy  $\pi$ . Assume all other aspects of the environment remain as described in the problem statement.

- (d) **True or False:** There is no constant value  $k \in \mathbb{R}$  such that  $R(s = \text{‘In Play’}, a = \text{‘Capture Golden Snitch’}) = k$  would lead to a different optimal policy  $\pi$ . Assume all other aspects of the environment remain as described in the problem statement.

- ☐ True  
☐ False

2. Formulate the task of deciding how long to study for each of our two 10-701 exams this semester as a single reinforcement learning problem. Specifically, briefly describe in words the following components:

(a) **Short Answer:** The state space  $\mathcal{S}$ .

(b) **Short Answer:** The action space  $\mathcal{A}$ ; make sure to specify what actions can be taken in any given state.

(c) **Short Answer:** The reward function or distribution.

(d) **Short Answer:** The transition function or distribution.



3. **Math:** Consider the case of a stochastic policy  $\pi$ , such that  $\pi(a \mid s)$  is the probability of taking action  $a$  in state  $s$  under stochastic policy  $\pi$ . If the current state is  $s_t$ , write the expectation of  $r_{t+1}$  in terms of  $\pi(a \mid s)$  and  $p(s', r \mid s, a)$ , the probability of transitioning to state  $s'$  with reward  $r$ , from state  $s$  and action  $a$ .

## 7 Ensemble Methods

1. In class, we set  $\alpha_t$  to  $\frac{1}{2} \ln \left( \frac{1-\epsilon_t}{\epsilon_t} \right)$  for AdaBoost. In this question, we will show that by setting  $\alpha_t$  to this value, the binary classification error approaches 0 as the number of weak learners approach  $\infty$ . Consider a binary classification problem where  $y_i \in \{-1, 1\}$ .

- (a) Show that

$$Z_t = (1 - \epsilon_t)e^{-\alpha_t} + \epsilon_t e^{\alpha_t}$$

- (b) Using what we set  $\alpha_t$  and the result from the previous part, show that

$$Z_t = 2\sqrt{\epsilon_t(1 - \epsilon_t)}$$

- (c) Show that the binary classification error is bounded by the exponential loss

$$\frac{1}{N} \sum_{i=1}^N \mathbb{1}(\text{sign}(h(x_i)) \neq y_i) \leq \frac{1}{N} \sum_{i=1}^N e^{(-y_i h(x_i))}$$

- (d) Given

$$\frac{1}{N} \sum_{i=1}^N e^{-y_i h_T(x_i)} = \prod_{t=1}^T Z_t$$

Show that

$$\frac{1}{N} \sum_{i=1}^N \mathbb{1}(\text{sign}(h(x_i)) \neq y_i) \rightarrow 0 \text{ as } T \rightarrow \infty$$

in other words, the binary classification error approaches 0 as the number of weak learners approaches  $\infty$

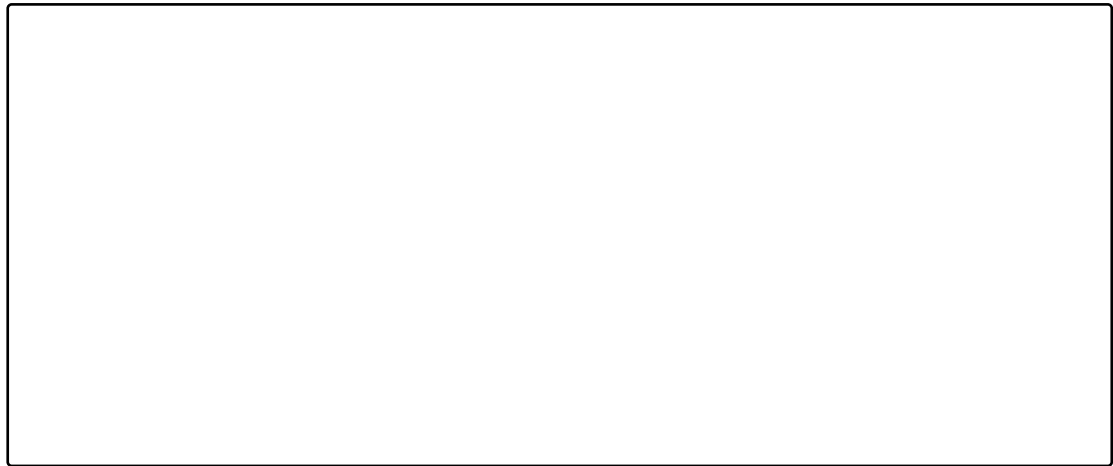
- (e) What condition does  $\epsilon_t$  need to satisfy in order to guarantee the previous part? Give an interpretation of this condition.

2. **Short Answer:** Suppose you train an Adaboost method and achieve 0 training error. Does this necessarily mean you should stop training? Briefly explain why or why not.

## 8 SVMs

1. One practical issue that arises in real-world classification tasks is that of imbalanced data, i.e., when one class is far more prevalent than all other classes. Consider, for example, a binary classification task with  $y \in \{-1, +1\}$  where 99% of all possible data points have label  $-1$ : a simple classifier that always predicts  $-1$  will achieve an accuracy of 99%. This can make it difficult to improve on/assess whether a particular machine learning model has learned anything at all! One way of learning in this setting is to penalize errors on the rare class more than errors on the common class.

- (a) **Math:** You decide to apply SVMs to the setting described above. Formulate a *primal* optimization problem that defines a soft-margin SVM where the soft error on training data points with label  $+1$  is penalized 50 times more than the soft error on training data points with label  $-1$  (**Hint:** the constraints associated with your optimization problem should be the same as the one presented in lecture, only the objective function will differ).



2. **Short Answer:** Explain in one sentence why we prefer the SVM dual form over the primal form?



## 9 Active Learning

1. **True or False:** For the Active SVM Algorithm, the algorithm should always request the label of the example farthest from the current separator.  
☐ True  
☐ False
2. In this problem you will reason about the *version space* and *region of disagreement* for the class of rectangles in the 2D dataset shown in Figure 2. Circles marked with ‘+’ and ‘-’ are positive and negative examples, respectively, while the unmarked circles are examples whose label we haven’t queried. Let  $\mathcal{H}$  be the hypothesis class of rectangles that predict positive for points inside and negative for points outside. Assuming we are in the realizable setting, please answer the following questions:
  - (a) On Figure 2 draw the smallest and largest rectangles that correctly classify the labeled examples. In 1–2 sentences, describe the version space in terms of these two rectangles.
  - (b) On Figure 2, mark each point in the region of disagreement with the letter ‘d’. For all other points, mark them with their correct label.
  - (c) Now suppose we query the label of the bolded point in Figure 3 and learn that it has a positive label. On Figure 3 draw the smallest and largest rectangles that correctly classify the labeled data, mark each point in the region of disagreement with a ‘d’, and mark all other points with their correct label.

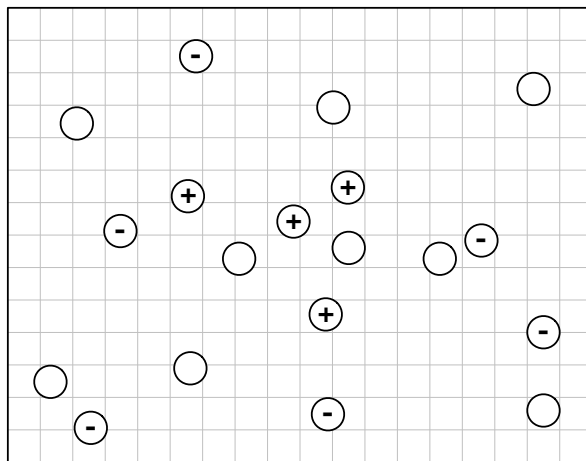


Figure 2: Dataset for parts (a) and (b).

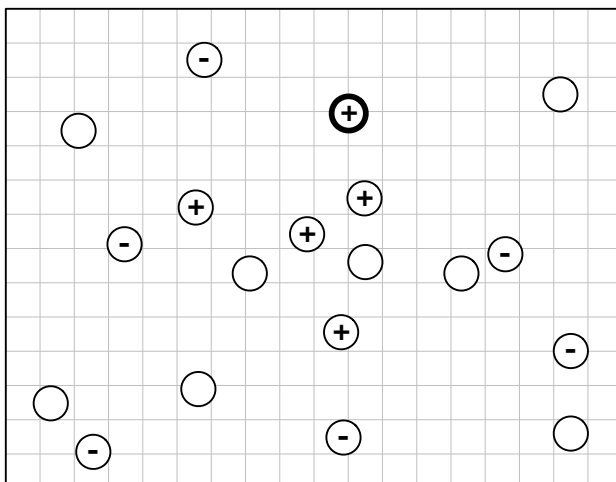


Figure 3: Dataset for part (c).

3. Answer the following True or False Statements. No justification is necessary:

- (a) [2 pts] Assume the problem is realizable. Consider a concept class  $H$  and a sequence of training instances  $\{(x_t, y_t)\}_{t=1}^n$ . Let  $H_t \subseteq H$  be the version space after the learner has received the  $t$ -th instance. Then  $H_t \neq \emptyset$ , for all  $t = 1, \dots, n$ .

- ☐ True  
☐ False

- (b) [2 pts] Assume the problem is realizable. Consider a concept class  $H$  and a sequence of training instances  $\{(x_t, y_t)\}_{t=1}^n$ . Let  $H_t \subseteq H$  be the version space after the learner has received the  $t$ -th instance. After all  $n$  examples have been observed, we have  $|H_n| = 1$ .

- ☐ True  
☐ False

- (c) [2 pts] Let  $X = \{(x_t, y_t)\}_{t=1}^n$  be a set of  $n$  training instances and  $H$  be the hypothesis space. Assume our current version space is  $H' \subseteq H$ . Pick another new sample  $(x_{t+1}, y_{t+1})$ . Let  $\hat{H} \subseteq H$  be the version space of  $X$  after querying  $x_{t+1}$ , i.e., the version space of  $X \cup \{(x_{t+1}, y_{t+1})\}$ . Then it must be that  $\hat{H} \neq H'$ .

- ☐ True  
☐ False

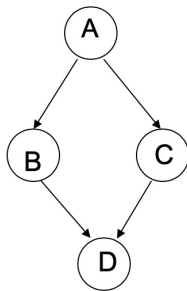
- (d) [2 pts] In the agnostic case, it is possible that none of the version spaces in the hypothesis space  $H$  are consistent with all the training instances  $\{(x_t, y_t)\}_{t=1}^n$ .

- ☐ True  
☐ False

## 10 Graphical Models

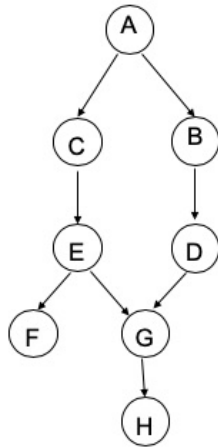
### 10.1 Graphical Models: Representation

1. **True or False?.** If  $X$  is conditionally independent of  $Y$  given  $Z$ , then  $Y$  is conditionally independent of  $X$  given  $Z$ . (in other words,  $X \perp Y|Z$  implies  $Y \perp X|Z$ ).
  - ☐ True
  - ☐ False
2. **Written answer:** Write down the product of terms that represents  $P(A, B, C, D)$  for this Bayes Network.



3. **Multiple choice:** The above Bayes Network implies certain conditional independencies. How many edges would this Bayes Network have if you modified it to imply *no* conditional independencies exist among its 4 variables?
  - ☐ 5
  - ☐ 6
  - ☐ 7
  - ☐ 8
  - ☐ none of the above

4. **Check all that apply:** Which of the following conditional independence relationships hold for this Bayes Net?

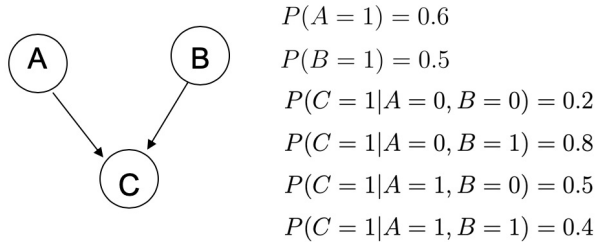


- ☐  $C \perp B | A$
- ☐  $A \perp G | D$
- ☐  $A \perp G | \{D, E\}$
- ☐  $A \perp D | \{B, G\}$
- ☐  $E \perp B | \{\}$
- ☐  $E \perp B | A$



## 10.2 Graphical Models: Inference

Consider the following Bayes Net, with the associated conditional probability distributions for each node, given its immediate parents.



Give the value of each of the following probabilities, rounding off to two decimal points (e.g., 0.29). [Hint1: some of these are easier than they look.] [Hint2: keep in mind the definition of conditional probability:  $P(X|Y) = P(X, Y)/P(Y)$ ]:

1. Answer to two decimal points:  $P(A = 1, B = 1, C = 1)$

2. Answer to two decimal points:  $P(C = 0|A = 1, B = 0)$

3. Answer to two decimal points:  $P(C = 1|A = 0)$

4. Answer to two decimal points:  $P(B = 1|A = 0, C = 1)$

5. Answer to two decimal points:  $P(C = 1)$